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Large-scale CSF proteome profiling identifies biomarkers for accurate diagnosis of frontotemporal dementia

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Abstract

Background Diagnosis of Frontotemporal dementia (FTD) and its specific underlying neuropathologies (frontotemporal lobar degeneration; FTLD-Tau and FTLD-TDP) are challenging, and thus, fluid biomarkers are needed to improve diagnostic accuracy.

Methods We used proximity extension assays to analyze 665 proteins in cerebrospinal fluid (CSF) samples from a multicenter cohort, which included patients with FTD ($n = 189$), Alzheimer's Disease dementia (AD; $n = 232$), and cognitively unimpaired individuals ($n = 196$). In a subset, FTLD neuropathology was determined based on phenotype or genotype (FTLD-Tau = 87 and FTLD-TDP = 67). Differences in protein expression profiles were analyzed using nested linear models. Penalized generalized linear modeling was used to identify classification protein panels, which were translated to custom multiplex assays and validated in two clinical cohorts (cohort 1: $n = 161$; cohort 2: $n = 162$), one autopsy-confirmed cohort ($n = 100$), and one genetic cohort ($n = 55$).

Results Forty-three proteins were differentially regulated in FTD compared to controls and AD, reflecting axon development, regulation of synapse assembly, and cell-cell adhesion mediator activity pathways. Classification analysis identified a 14- and 13-CSF protein panel that discriminated FTD from controls (FTD diagnostic panel, AUC: 0.96) or AD (FTD differential diagnostic panel, AUC: 0.91). Custom multiplex panels confirmed the strong discriminative performance between FTD and controls (AUCs > 0.96) and between FTD and AD (AUCs > 0.88) across three validation cohorts, including one with autopsy confirmation (AUCs > 0.90). Validation in genetic FTD (including *C9orf72*, *GRN*, and *MAPT* mutation carriers) revealed high accuracy of the FTD diagnostic panel in identifying both the presymptomatic (AUCs > 0.95) and symptomatic (AUC: 1) stages. Six proteins were differentially regulated between FTLD-TDP and FTLD-Tau. However, a reproducible classification model could not be generated (AUC: 0.80).

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Conclusions Overall, this study introduces novel FTD-specific biomarker panels with potential use in diagnostic settings.

Keywords FTD, FTLD, Tau, TDP43, Biomarkers, Proteomics, CSF

Background

Frontotemporal dementia (FTD) is the second most common form of young-onset dementia (i.e., defined as dementia with symptom onset before the age of 65). FTD presents heterogeneously, comprising various clinical, neuropathological, and genetic forms. From a clinical perspective, FTD can present with either behavioral and social changes (behavioral variant FTD, bvFTD), language impairment (semantic variant primary progressive aphasia, svPPA; and non-fluent variant PPA, nfvPPA), or motor dysfunction (corticobasal syndrome, CBS; and progressive supranuclear palsy, PSP) [1–4]. FTD is caused by different underlying neuropathologies leading to frontal temporal lobar degeneration (FTLD). Approximately 50% of FTLD cases develop aggregates of the microtubule Tau protein (FTLD-Tau), while 45% are characterized by cytoplasmic inclusion of the TDP-43 protein (FTLD-TDP). A small percentage of FTLD cases (5%) develop aggregates of the FUS protein (FTLD-FET) [5–7]. These pathologies can be predicted in familial forms of FTLD, where autosomal dominant mutations in the microtubule-associated protein tau (*MAPT*) gene lead to FTLD-Tau pathology, while mutations in the progranulin (*GRN*) or chromosome 9 open reading frame 72 (*C9orf72*) genes are associated with FTLD-TDP pathology [5, 8]. However, genetic FTD accounts for only 10–20% of FTLD cases, and in many families, the mutation status remains unknown, and novel genetic variants continue to be identified [9–12]. In sporadic cases, which are more common, the different neuropathological subtypes poorly correlate with the clinical presentation, with the exception of FTD accompanied with motor neuron disease and svPPA that shows a clinicopathological correlation with TDP-43 [13]. Thus, the different clinical phenotypes can have overlapping pathologies despite diverse genetic backgrounds, stressing the complexity of the disease and the importance of developing pathology-specific biomarkers, which are needed for targeted treatment.

Currently, there are no fluid biomarkers that specifically capture underlying FTD pathogenesis [14]. Given the overlapping clinical features with Alzheimer's disease (AD) [15, 16], the analysis of the core AD-CSF biomarkers (i.e., amyloid-beta 1–40 and 1–42, A β 1–40 and A β 1–42 or the A β 1–42/A β 1–40 ratio; phosphorylated Tau181, pTau181; and total Tau, tTau) is often used to exclude AD diagnosis [17]. The concentrations of Neurofilament light

(NfL) in CSF and blood are strongly increased in FTD but also in other neurodegenerative disorders since it reflects neuroaxonal damage [18, 19]. Nevertheless, this biomarker is considered valuable for distinguishing FTD from non-neurodegenerative disorders, such as primary psychiatric diseases, and it also shows good prognostic potential [20, 21]. Several studies have shown that the CSF pTau/tTau ratio could demarcate the main FTLD subtypes but with limited diagnostic accuracy [22, 23]. However, promising findings have been recently obtained with plasma GFAP/NfL ratio (FTLD-Tau vs. FTLD-TDP; AUC: 0.89), which requires additional validation in external cohorts [24]. Overall, there is an unmet need for fluid biomarkers specifically identifying FTD and its underlying pathologies, which is essential for accurate diagnosis, clinical trial inclusion, and to monitor the effects of treatments [25].

CSF proteomics offer the opportunity to study brain changes and associated processes in vivo. As extensively explored in AD, such analysis could offer insights into the disease's etiology and reveal novel biomarker candidates [26–29]. Previous FTD biomarker discovery studies using unbiased mass spectrometry (MS)-based technologies identified multiple biomarker candidates (e.g., YKL-40 in FTD and NPTXR in *GRN* mutation carriers) [30, 31]. However, subsequent validation efforts did not show sufficient diagnostic accuracy of these markers [32–35]. The clinical and pathological heterogeneity of FTD and low sample sizes could explain the lack of validated biomarkers [30, 31]. In addition, the translation of novel FTD biomarkers could be hampered due to the use of different technologies between the discovery (e.g., MS) and validation (immunoassay-based methods) phases. Therefore, we followed a similar strategy to that previously applied for AD and dementia with Lewy bodies (DLB) [29, 36]. We here used the high-throughput immune-based proximity extension assay (PEA) to analyze the CSF proteome of an extensive and well-characterized cohort of patients with FTD presenting at memory clinics. We first aimed to identify CSF protein changes specifically associated with FTD and its main pathological subtypes (i.e., FTLD-Tau and FTLD-TDP). Next, we aimed to translate these findings into clinically feasible CSF protein panels to discriminate FTD and its pathological subtypes. Finally, we validated these panels in four independent cohorts, including an autopsy-confirmed and a genetic cohort.

Method

Participants

As part of our previous work [29], the discovery cohort (total = 617; Table 1) included CSF samples from patients diagnosed with FTD ($n = 189$), AD dementia ($n = 232$), and cognitively normal individuals (CON, $n = 196$). Most of the samples were selected from the Amsterdam Dementia Cohort (ADC; 120 FTD, 214 AD, 190 CON) [37]. To enrich for samples from patients with confirmed FTD, additional cases from the Center for Neurodegenerative Disease Research at the University of Pennsylvania (Penn; 46 FTD, 18 AD, 6 CON), Erasmus Medical Center (19 FTD), and the Goizueta Alzheimer's Disease Research Center at Emory University (4 FTD) were selected. For a subset of the FTD patients (82%), the underlying neuropathology was known or could be predicted based on specific clinical diagnosis (FTLD-Tau = 87, FTLD-TDP = 67). FTLD-Tau was confirmed based on autopsy ($n = 17$), *MAPT* mutation ($n = 16$), and further enriched with patients clinically diagnosed with PSP ($n = 31$) and CBS ($n = 23$), which primarily associates with Tau neuropathology [38]. The FTLD-TDP group included autopsy-confirmed cases ($n = 25$) and patients with *GRN* ($n = 8$) or *C9orf72* ($n = 23$) mutations, and further enriched with patients clinically diagnosed with svPPA ($n = 11$), which have a high likelihood of having TDP pathology [13]. Four additional validation cohorts were included for validation of the custom panels: two clinical cohorts, from the ADC (51 FTD, 55 AD, and 55 controls) and Sant Pau Initiative on Neurodegeneration (SPIN; 54 FTD, 53 AD, and 55 controls) [39], one FTLD/AD autopsy confirmed cohort from the Reference Center for Biological Markers of Dementia (BIODEM) and the neurobiobank of the Institute Born-Bunge (IBB) from the University of Antwerp (UAntwerp) (BIODEM-UAntwerp; 41 autopsy-FTLD; aFTLD, including 7 aFTLD-Tau and 31 aFTLD-TDP, 3 aFTLD-UPS, and 30 autopsy-AD; aAD). Noteworthy, 29 cognitively unimpaired controls from BIODEM-UAntwerp were included in this cohort, though not autopsy-confirmed. Lastly, individuals from the Frontotemporal Dementia Risk Cohort (FTD RisC) at Erasmus Medical Center were included. This cohort comprises patients with FTD caused by autosomal dominant mutations (14 symptomatic mutation carriers, including 10 *C9orf72*, 3 *GRN*, and 1 *MAPT*). Additionally, their at-risk relatives were also included, consisting of 8 non-carriers and 33 presymptomatic mutation carriers (22 *C9orf72*, 7 *GRN*, and 4 *MAPT*).

All participants underwent standard neurological screening and cognitive testing. FTD and AD diagnoses were made according to international consensus criteria [1–4, 40]. The autopsy-confirmed cohort included cases with a definite diagnosis according to international neuropathological examination guidelines for FTLD [41] and AD [42]. The genetic FTD cohort included participants identified as symptomatic based on international guidelines [1, 2]. Symptomatic onset was defined as the point at which caregivers first reported the presence of symptoms. Mini-mental state examination (MMSE) was used as a measurement for global cognition in all groups. In addition, for a subset of FTD cases from the ADC (discovery cohort, $n = 62$; clinical cohort 1, $n = 44$), the FTLD CDR* plus NACC score was used as a clinical measure specific for FTD disease severity [43]. The control group included individuals with subjective cognitive decline, who scored normally on cognitive examinations with a negative AD CSF biomarkers profile.

CSF was collected and biobanked according to established protocols [44]. Concentrations of CSF A β 42, pTau, and tTau were used to support AD diagnosis and to evaluate the presence of AD co-pathology in the FTD cases. These were analyzed locally as part of the diagnostic work-up using commercially available kits (ADC, Erasmus MC, and BIODEM-UAntwerp: ELISA Innostest A β (1–42), hTAUAg, pTau (181P; Fujirebio, Ghent), or ADC: Elecsys A β 42, tTau and pTau (181P) CSF assays (Roche Diagnostics); Penn and Emory: Luminex xMAP INNO-BIA AlzBio3 (Luminex, Bio-technique); SPIN: Lumipulse G600, Fujirebio). Predefined cut-offs were used to define a positive AD biomarker profile. (ADC: tTau/A β 42 ratio > 0.46; Penn: tTau/A β 42 ratio > 0.30; Emory: A β 42/tTau ratio < 6; SPIN: A β 42/A β 40 ratio < 0.062, tTau > 456 pg/mL and pTau > 63 pg/mL). In the ADC, Innostest A β 42 concentrations were adjusted for drift over time [45]. CSF biomarkers measured on Luminex were transformed to Innostest values using conversion formulas based on Passing-Bablok regression analysis using cases for which both Luminex and Innostest values were available, as described previously [29]. CSF NfL was measured in a subset of cases from the ADC (discovery cohort: 42 CON, 74 FTD, and 54 AD; clinical cohort 1: 8 CON, 19 FTD, and 20 AD) either by NF-light® ELISA (Uman Diagnostics, Sweden) or with the single molecular array (Simoa®) NF-light™ advantage kit (Quanterix, USA) [46]. NfL concentrations measured by ELISA were converted to Simoa values using Passing-Bablok regression analysis, as described previously [46].

Table 1 Cohort characteristics

	Discovery Cohort			Validation Cohort 1			Validation Cohort 2			FTLD/AD neuropathological		
	Control	FTD	AD	Control	FTD	AD	Control	FTD	AD	Control	aFTLD	aAD
n	196	189	232	55	51	55	55	54	53	29	41	30
FTD Clinical Phenotype		35 bvFTD 23 CBS 31 PSP 11 svPPA 0 rfvPPA			38 bvFTD 0 CBS 0 PSP 5 svPPA 2 rfvPPA			31 bvFTD 18 CBS-PSP* 1 lvPPA 2 svPPA 2 rfvPPA			na na na na na	
		42 Autopsy-confirmed 47 Mutation Carriers			0 Autopsy-confirmed 6 Mutation Carrier			0 Autopsy-confirmed 0 Mutation Carriers			41 Autopsy-confirmed na	
Sex, Female n (%)	72 (36.7)		96 (41)	22 (40)	17 (33)	23 (42)		21 (38)	34 (64)	13 (45)	20 (49)	15 (50)
Age	58 (53 - 63) ^{ab}	65 (59 - 70) ^c	66 (59 - 72) ^c	58 (56 - 62) ^{ab}	63 (59 - 69) ^c	66 (62 - 70) ^c		76 (71 - 79) ^c	74 (68 - 76) ^c	64 (61 - 66) ^b	64 (55 - 71) ^b	72 (64 - 76) ^{ac}
MMSE	28 (27 - 29) ^{ab}	25 (22 - 27) ^{bc}	21 (17 - 24) ^{ac}	29 (27 - 30) ^{ab}	25 (23 - 27) ^{bc}	20 (18 - 23) ^{ac}		29 (29 - 30) ^{ab}	22 (19 - 25) ^c	29.5 (29 - 30)	20 (14 - 26)	14 (7 - 21)
CSF Aβ42, pg/mL	1120 (1023 - 1241) ^{ab}	985 (782 - 1193) ^{bc}	602 (535 - 657) ^{ac}	1283 (1085 - 1373) ^{ab}	1086 (888 - 1253) ^{bc}	566 (525 - 661) ^{ac}		1194 (899 - 1397) ^{ab}	537 (441 - 646) ^{ac}	1108 (952 - 1216) ^{ab}	681 (494 - 866) ^{bc}	404 (306 - 497) ^{ac}
CSF pTau, pg/mL	38 (31 - 45) ^{ab}	44 (36 - 57) ^{bc}	92 (77 - 112) ^{ac}	54 (39 - 66) ^c	46 (39 - 61) ^b	65 (50 - 92) ^a		38 (28 - 47) ^b	117 (81 - 165) ^{ac}	47 (39 - 56) ^b	39 (29 - 49) ^b	72 (49 - 84) ^{ac}
CSF Ttau, pg/mL	213 (167 - 257) ^{ab}	324 (239 - 414) ^{bc}	751 (581 - 1005) ^{ac}	292 (213 - 443) ^b	362 (254 - 487) ^b	479 (343 - 684) ^a		324 (224 - 434) ^b	740 (546 - 976) ^{ac}	203 (168 - 257) ^{ab}	330 (222 - 441) ^{bc}	535 (325 - 674) ^{ac}
APOE4 carrier, n (%)	48 (25)	46 (24)	132 (57)	15 (27)	13 (26)	39 (71)		8 (15)	22 (41.5)	na	na	na
FTLD-Subtypes		FTLD-Tau (87 total): 17 Autopsy-confirmed 16 MAPT 31 PSP 23 CBS FTLD-TDP (67 total): 25 Autopsy-confirmed 8 GRN 23 C9orf72 11 svPPA			na			na			aFTLD-Tau (7 total) aFTLD-TDP (31 total) aFTLD-UPS (3 total)	

Continuous variables are presented as median ± interquartile range and dichotomous data as the number of cases with a percentage of the total (%)

Differences between groups were determined using the Kruskal-Wallis test with Bonferroni correction or Chi-squared test

Analysis of covariance was performed for CSF biomarker analysis, adjusting for age and sex when appropriate

In the discovery cohort, biomarker values from the luminex were transformed to innote values using passingbablok regression formulas (method)

For validation cohort 1, the first set of biomarker values corresponds to data obtained using Innote, while the second set corresponds to data obtained using Elecsys

CSF Biomarker data from validation cohort 2 were obtained with Lumipulse

a, p<0.05 compared to FT(L)D

b, p<0.05 compared to AD

c, p<0.05 compared to CON

* 18 Patients were on the spectrum of CBS and PSP, with 11 showing a clinical profile more consistent with CBS, and 4 exhibiting features more indicative of PSP

Abbreviations: FTD = frontotemporal dementia, AD = Alzheimer's Disease, MMSE = mini-mental state examination, CSF = cerebrospinal fluid,

FTLD = frontotemporal lobar degeneration, na = not applicable, TDP = Transactive response DNA binding protein of 43, FUS = fused in sarcoma, aFTLD = autopsy confirmed FTLD, aAD = autopsy confirmed AD

CSF protein profiling

979 CSF proteins were quantified using 11 Olink Target 96 validated multiplex panels based on proximity extension assay (PEA) technology (Olink Proteomics, Uppsala, Sweden) that were available at the time of analysis (Cardiometabolic, Cardiovascular II and III, cell regulation, development, immune response, inflammation, metabolism, neurology, oncology II and organ damage). CSF samples were randomized across multiple plates containing intra- and inter-plate quality controls (QCs) from the manufacturer and measured in two different rounds. Each round of measurement contained 16 bridging samples covering different clinical groups, which were used as a reference to account for potential batch effects. For each protein, the lower limit of detection (LOD) was determined by the company and defined as three standard deviations above the background from the negative controls included on every plate. Proteins were excluded from further analysis if their levels were below the LOD in more than 15% of the samples. A total of 665 proteins (642 unique proteins) were ultimately included for statistical analysis of the discovery cohort, similar to our previous study [29]. Protein abundance was reported in normalized protein expression (NPX) values.

Development of custom PEA assays

Multiplex-PEA assays were custom-developed by the manufacturer following standardized protocols [47]. We developed assays to measure 18 out of 24 proteins selected by the classification analysis described below. CSF samples from validation cohorts were randomized across plates. Each plate additionally included: four CSF QC samples, a negative control, and three calibrators used for normalization. Each custom assay had a LOD determined by the company, defined as three standard deviations above the background from the negative controls. Precision (intra- and inter-assay CV) was calculated using the four CSF QC samples (Supplementary Table 1). No cross-reactivity between assays for the specific proteins was detected. Samples from the validation cohorts were randomized across plates and normalized for any plate effects using the built-in inter-plate controls according to the manufacturer's recommendations. Protein levels are reported in NPX values.

Statistical analyses

All processing and statistical analysis were performed in R version 4.2.1. Baseline demographics were tested by Kruskal-Wallis test followed by Bonferroni post-hoc, or Pearson's chi-square test for continuous and categorical variables, respectively. Differences in protein abundance of the CSF proteome data was tested with nested linear models including age and sex in the model for the comparison between different groups (FTD vs. controls, FTD

vs. AD, FTLD-Tau vs. controls, FTLD-TDP vs. controls, and FTLD-Tau vs. FTLD-TDP) as previously performed [29, 36]. For each pairwise comparison, multiplicity was taken into account by controlling the False Discovery Rate (FDR) at $q \leq 0.05$ based on the number of features analyzed [48].

We next evaluated which CSF protein combination (CSF panels) could best discriminate the groups of interest while keeping the number of markers to a minimum so that they can be ultimately translated into small, practical custom panels, as described previously [29]. For this purpose, binary classification signatures (FTD vs. controls: FTD Diagnostic Panel, FTD vs. AD: FTD Differential Diagnostic Panel, and FTLD-Tau vs. FTLD-TDP: FTLD Subtype Panel) were constructed by way of penalized generalized linear modeling (GLM) with an elastic net penalty (a linear combination of lasso and ridge penalties) in the discovery cohort using the glmnet package, including age and sex as covariates in the model [29, 49, 50]. This penalty enables estimation in settings where the feature-to-sample ratio is too high for standard generalized linear regression. Moreover, it performs automatic feature decorrelation as well as feature-selection. For each classification exercise, we compare multiple models that reflect (a) a grid of values for the elastic-net mixing parameter, reflecting strong decorrelation to a pure logistic lasso regression, and (b) a grid of values reflecting the maximum number of proteins that may be selected under each model (21 markers maximum). The former grid (a) considers that we have little information on the collinearity burden in the data. The latter grid (b) considers that we want to keep the number of selected proteins relatively low for the future development of customized panels. The optimal penalty parameters in the penalized models were determined based on (balanced) 10-fold cross-validation of the model likelihood [29, 49]. The cross-validation was performed with balanced folds, by which each fold has an outcome group ratio close to the corresponding ratio in the full data set, also referred to as stratified cross-validation. The predictive performance of all models was assessed by way of (the comparison of) Receiver Operating Characteristic (ROC) curves and Area Under the ROC Curves (AUCs). The model with the highest AUC and lowest number of markers for each classification signature was selected. The fold-based selection proportions for each marker were assessed to identify and select the most promising markers within each model (i.e., features that are stably selected across each individual fold, thereby minimizing potential overfitting). To reflect the manual selection pressure for these final marker sets, each final logistic signature was subjected to a ridge-regularization with a penalty parameter of 0.1. The performance (AUC) was evaluated by internal validation: repeated 5-fold cross-validation with 1000

repeats. The 95% confidence interval around the resulting AUCs was based on resampling quantiles (percentile method). External validations assessed the performance of the final models with the markers of interest in the validation cohorts. We additionally compared our identified classification models to CSF NfL to discriminate FTD from controls and AD in the subset of cases for which this information was available.

Functional enrichment analysis was performed for all comparisons described above using Metascape, selecting GO biological Processes as an ontology source [51]. The total number of CSF proteins optimally analyzed was included as the enrichment background ($n=642$). Default parameters were used for the analysis in which terms with a p -value <0.01 , a minimum count of 3, and an enrichment factor >1.5 were collected and grouped into clusters based on their membership similarities.

Partial non-parametric correlation analysis was performed to understand the association among proteins within the CSF panels with cognitive function (MMSE score) and disease severity (FTLD-CDR⁺ plus NACC scores) [43]. This analysis was corrected for age, sex, and the clinician who performed the FTLD-CDR examination (included as a dummy variable), and was conducted both in the total discovery cohort and specifically within the FTD group.

Results

Subjects

An overview of the study is presented in Fig. 1A. The discovery cohort included a total of 617 samples, all measured as part of previous CSF proteomic studies [29, 36]. Custom multiplex panels were developed and validated in two independent clinical cohorts (validation cohort 1: $n=161$; validation cohort 2: $n=162$), one FTLD/AD autopsy-confirmed cohort ($n=100$), and one genetic FTD cohort ($n=55$). The demographic characteristics for the FT(L)D and AD cohorts are described in Table 1 and the genetic FTD cohort is described in Supplementary Table 1. Controls were younger compared to other diagnostic groups in the discovery cohort, clinical cohort 1, and clinical cohort 2. Patients with AD were older, and patients with FTLD were overall younger in the FTLD/AD autopsy cohort compared to the other cohorts. In the genetic FTD cohort, presymptomatic individuals were significantly younger compared to the symptomatic group (Supplementary Table 1).

CSF proteins are differentially regulated in FTD compared to controls and AD

CSF proteome profiling revealed 92 proteins differentially regulated between FTD and controls, of which 42 proteins were decreased and 50 proteins were increased ($q<0.05$; Fig. 1B, Supplementary Table 2). The top 5

differentially regulated proteins related to FTD (median $q: 2.5^{-07}$) are involved in the regulation of the Wnt signaling pathway (WIF1), tissue remodeling (MMP7), neuronal biosynthesis (APP), cell proliferation (NPDC1), and innate immunity (IL1RL2). Sensitivity analysis showed that excluding FTD patients with AD co-pathology ($n=19$, 10%) had minimal effect on the CSF proteome profiling results (Supplementary Fig. 1). Next, we sought to determine whether the protein changes identified were especially related to FTD. We show that 36 proteins (40%) were uniquely dysregulated in FTD compared to controls and AD (e.g., WIF1, ROBO2, and SLITRK2; Fig. 1C). Nineteen proteins (20%) were dysregulated in FTD and AD compared to controls, which likely represent general dementia processes (e.g., CHIT1, MMP10, and MMP12; Fig. 1C). Twenty-eight proteins (30%) were dysregulated across all comparisons: controls versus FTD and AD, and FTD versus AD. Most of these proteins (21 out of 28; e.g., MIF, ITGB2, and sTREM1) were to some extent increased in FTD but were more prominently dysregulated in AD, indicating that they are likely more related to AD pathophysiological processes. However, 7 out of the 28 proteins (e.g., CHL1, GPC1, CNTN5, HBEGF, CLSTN2, GCP5) showed an opposite direction of change in FTD and AD and are likely differently involved in the pathophysiology of these types of dementias. Functional enrichment analysis of the proteins related to FTD pathophysiology (i.e., those that were increased in FTD as well as those that were specifically dysregulated in FTD compared to AD, $n=43$) revealed associations with axon development, regulation of synapse assembly, and cell-cell adhesion mediator activity (Fig. 1D).

CSF protein panels discriminate with high accuracy FTD from controls and AD

Next, we aimed to identify CSF biomarkers specific for FTD. One of the strongest proteins specifically dysregulated in FTD was WIF1, however, its performance as a single marker to discriminate FTD from controls was moderate (AUC: 0.794, 95% CI: 0.75–0.84). Thus, we next performed a classification analysis, followed by internal cross-validation, to investigate whether a combination of proteins could discriminate FTD from controls with higher accuracy (CSF panels, Fig. 1A). We identified a panel of 14 CSF proteins that discriminated FTD from controls with high accuracy (FTD diagnostic panel; AUC: 0.96, 95% CI: 0.91–0.99; Fig. 2A), which was comparable to the performance of CSF NfL (AUC: 0.96, 95% CI: 0.92–1; Fig. 2B). The model included proteins that were dysregulated specifically in FTD (i.e., WIF1, MMP7, GAL, VEGFA, NPDC1, APP, and CCL11; Supplementary Fig. 2), as well as proteins related to common neurodegenerative processes and other neurodegenerative dementias (i.e., MMP1, MMP10, CHIT1, CCL3, PRDX1,

and DDC; Supplementary Fig. 2), or reported to be associated with AD (i.e., CLEC5A; Supplementary Fig. 2). The performance of the FTD diagnostic panel to discriminate FTD from AD was considerably lower (FTD diagnostic panel AUC: 0.77, 95% CI: 0.67–0.86; for comparison: CSF NfL AUC: 0.80, 95% CI: 0.72–0.88, Fig. 2B). We thus next investigated whether we could find a combination of proteins that optimally discriminated FTD from AD. A panel consisting of 13 CSF proteins was identified that discriminated FTD from AD with high accuracy (FTD differential diagnostic panel; AUC: 0.91, 95% CI: 0.85–0.96; Fig. 2C). This panel contained proteins associated with FTD (i.e., GZMB, MMP7, CCL11, NPDC1, PLTP, and APEX1; Supplementary Fig. 3) but also proteins associated with AD (i.e., ABL1, ENO2, ITGB2, SMOC2,

and THOP1; Supplementary Fig. 3), as also reported in our previous AD focused PEA study [29]. The model also included two proteins with no differences across groups, likely due to the model's adjustment for inter-individual variability (i.e., VEGFR3 and KAZALD1; Supplementary Fig. 3) [52]. Of note, three proteins (i.e., MMP7, NPDC1, and CCL11) were included in both the FTD diagnostic and FTD differential diagnostic panel. The FTD-related proteins from these panels are associated with diverse biological pathways including inflammatory processes (GZMB), tissue remodeling (MMP1 and MMP7), cell proliferation (NPDC1), oxidative stress (APEX1), vascular functioning (VEGFA), and neuronal biosynthesis and functioning (APP and GAL) [53–57].

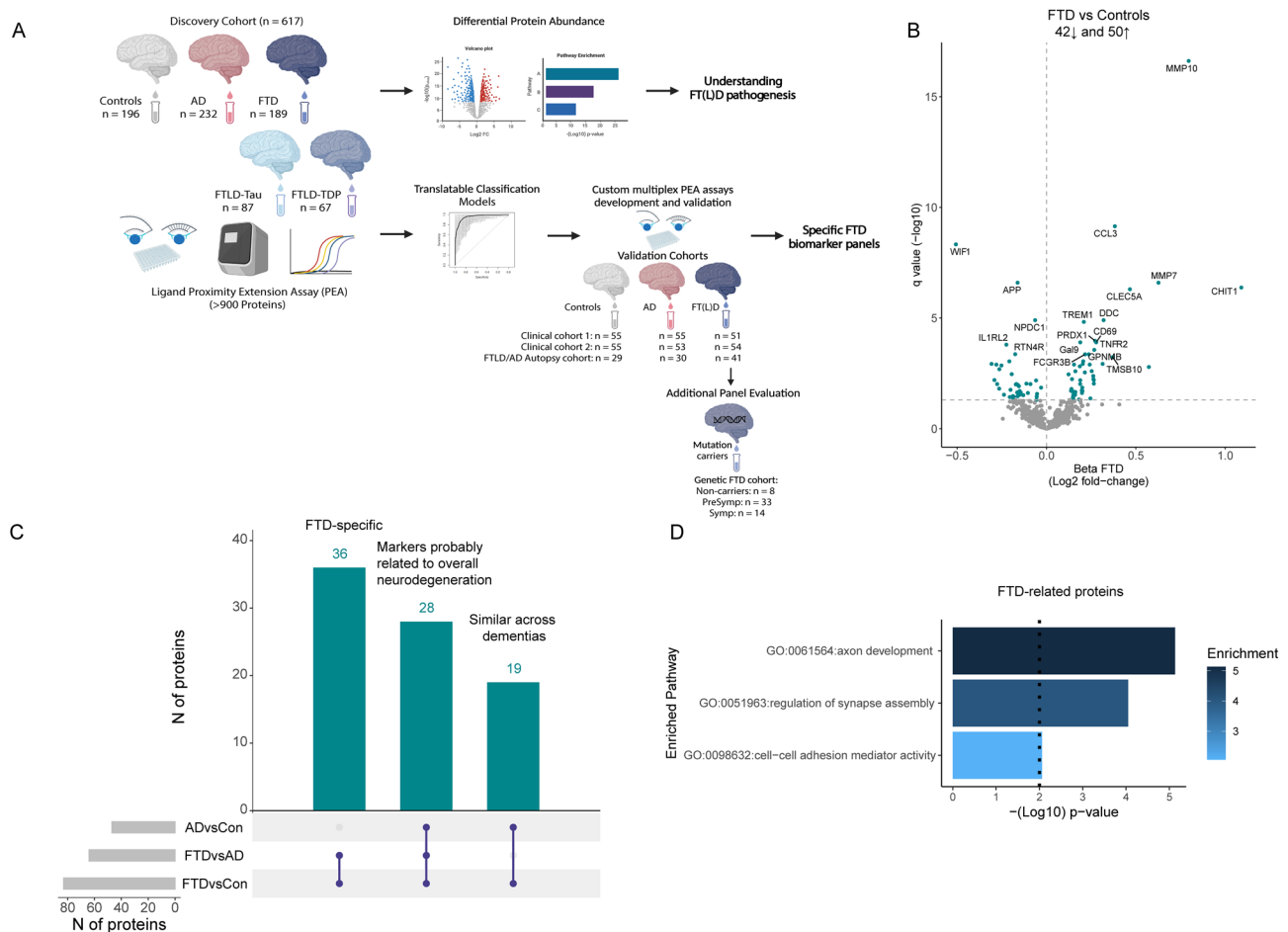


Fig. 1 Overview study design and differential abundance of CSF proteins in FTD. **(A)** More than 900 proteins were measured using the antibody-based PEA technology in CSF from 196 cognitively unimpaired controls (white), 189 FTD (blue), and 232 AD (red) patients. Differential protein abundance was investigated and classification models were constructed. Custom PEA assays using the proteins identified in the classification models were developed and validated in four independent cohorts, including an FTD/AD autopsy cohort and genetic cohort. **(B)** Volcano plots show that 93 CSF proteins were differentially regulated between FTD and controls. Each dot represents a protein. The beta coefficients (log2 fold-change) are plotted versus q values (-log10-transformed). Proteins significantly dysregulated after adjusting for false discovery rate (FDR, $q < 0.05$) are depicted in blue. The total number of proteins that are down-regulated ($n = 42$, left) or up-regulated ($n = 51$, right) are depicted. Horizontal dotted line indicates the significance threshold. **(C)** UpSet plot shows proteins dysregulated between FTD and controls and also dysregulated between FTD and AD or AD and controls. **(D)** Bar graphs depicting the biological pathways enriched in protein specifically dysregulated in FTD. The dotted line represents the significant threshold ($p < 0.01$). The corresponding GO number and biological process is depicted on the left side. Stronger colors represent higher significant enrichment

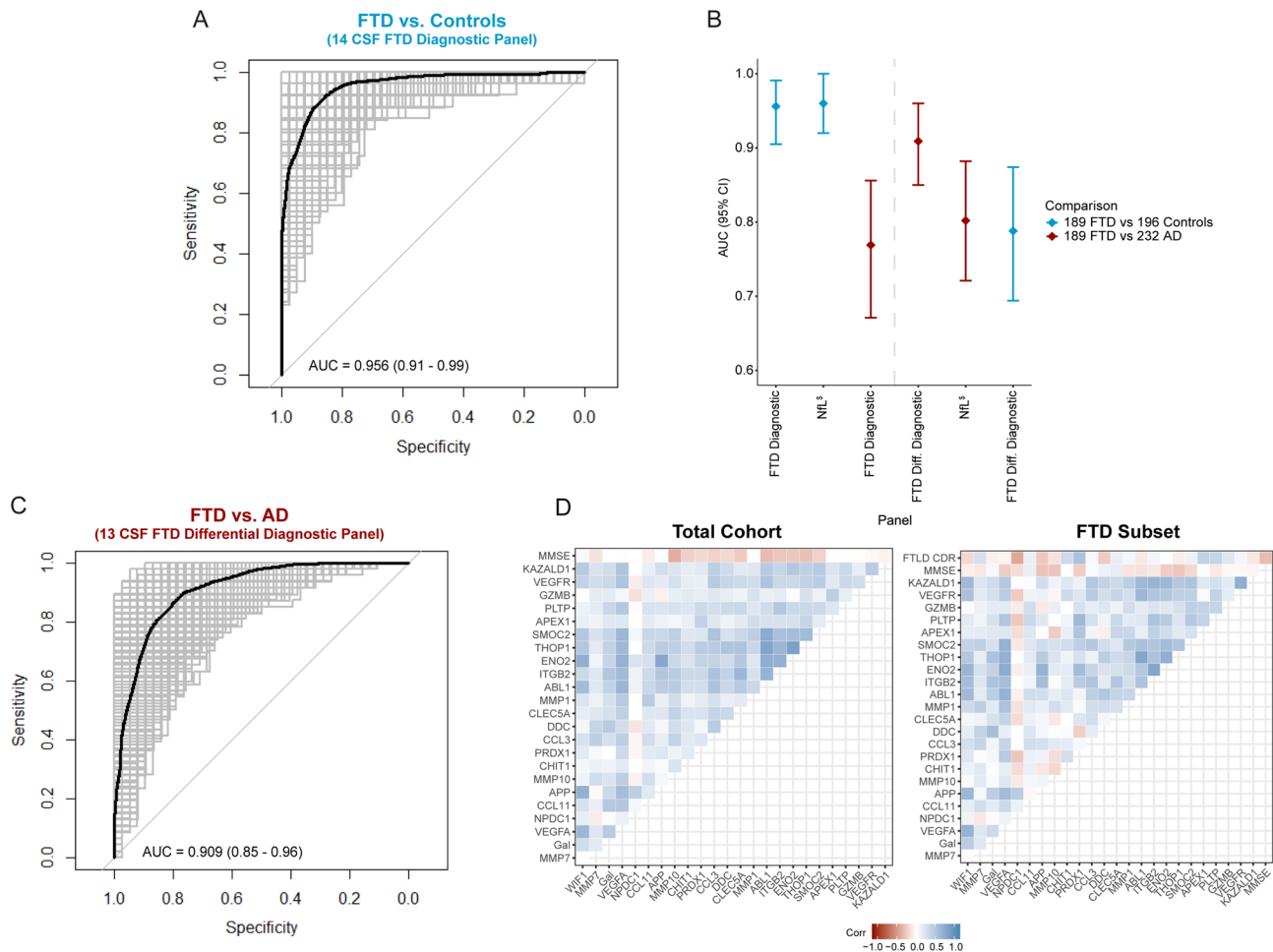


Fig. 2 CSF biomarker panels for the diagnosis of FTD. **(A)** Receiver operating characteristic (ROC) curves depict the performance of 14-CSF biomarker panel discriminating FTD ($n = 189$) from controls ($n = 196$). Black line is the mean area under the curve (AUC) overall re-samplings (1000 repeats of 5-fold cross-validation, grey lines). **(B)** Forest plot shows the different AUC and 95% confidence interval for the 14 and 13 CSF biomarker panels and NfL to discriminate between FTD and controls (blue) or AD (red). **(C)** ROC curves depict the performance of 13-CSF biomarker panel discriminating FTD ($n = 189$) from AD ($n = 232$). **(D)** Correlation matrix heatmap representing the Spearman's correlation coefficient in-between the proteins selected in each panel, MMSE score, and FTLD-CDR scores in the total cohort, and in a subset of the FTD group ($n = 62$)

Next, we sought to determine if the proteins included in our FTD panels are associated with disease severity and cognition as measured by FTLD-CDR and MMSE scores. In the FTD patients, we observed that PRDX1 ($Rho = 0.47$, $p < 0.001$) had a moderate to strong correlation while NPDC1 ($Rho = -0.37$, $p < 0.001$), APEX ($Rho = 0.35$, $p < 0.01$), PLTP ($Rho = 0.33$, $p < 0.01$), and CHIT1 ($Rho = 0.28$, $p < 0.05$) showed a moderate correlation with FTLD-CDR scores (Fig. 2D). In the total cohort, we show that the proteins associated with AD (ABL1, ENO2, ITGB2, SMOC2, and THOP1) together with MMP10 had the strongest correlations with MMSE scores (Rho 's between -0.19 and -0.34 , all $p < 0.001$; Fig. 2D).

CSF proteins are differentially regulated in FTLD-Tau or FTLD-TDP subtypes

Our second aim was to investigate which protein dysregulations characterized the FTLD-Tau and FTLD-TDP subtypes. Compared to controls, we observed 60 proteins dysregulated in FTLD-Tau (top 5: MMP10, DDC, CCL3, MMP7, and WIF1; median q 4.2×10^{-8} , $q < 0.05$; Fig. 3A, Supplementary Table 3) and 120 proteins that were dysregulated in FTLD-TDP (top 5: APP, NPDC1, WIF1, B4GAT1, and ROBO2; median q 1.2×10^{-6} ; Fig. 3B, Supplementary Table 4). When comparing FTLD-Tau to FTLD-TDP, we observed that 6 proteins (COCH, Siglec9, VSIG4, GRN, CD84, and C1QTNF1) were dysregulated between these groups ($q < 0.05$; Fig. 3C, Supplementary Table 5). This number increased to 185 dysregulated proteins when nominal significance was considered (i.e., $p < 0.05$, Supplementary Table 5). To show which proteins

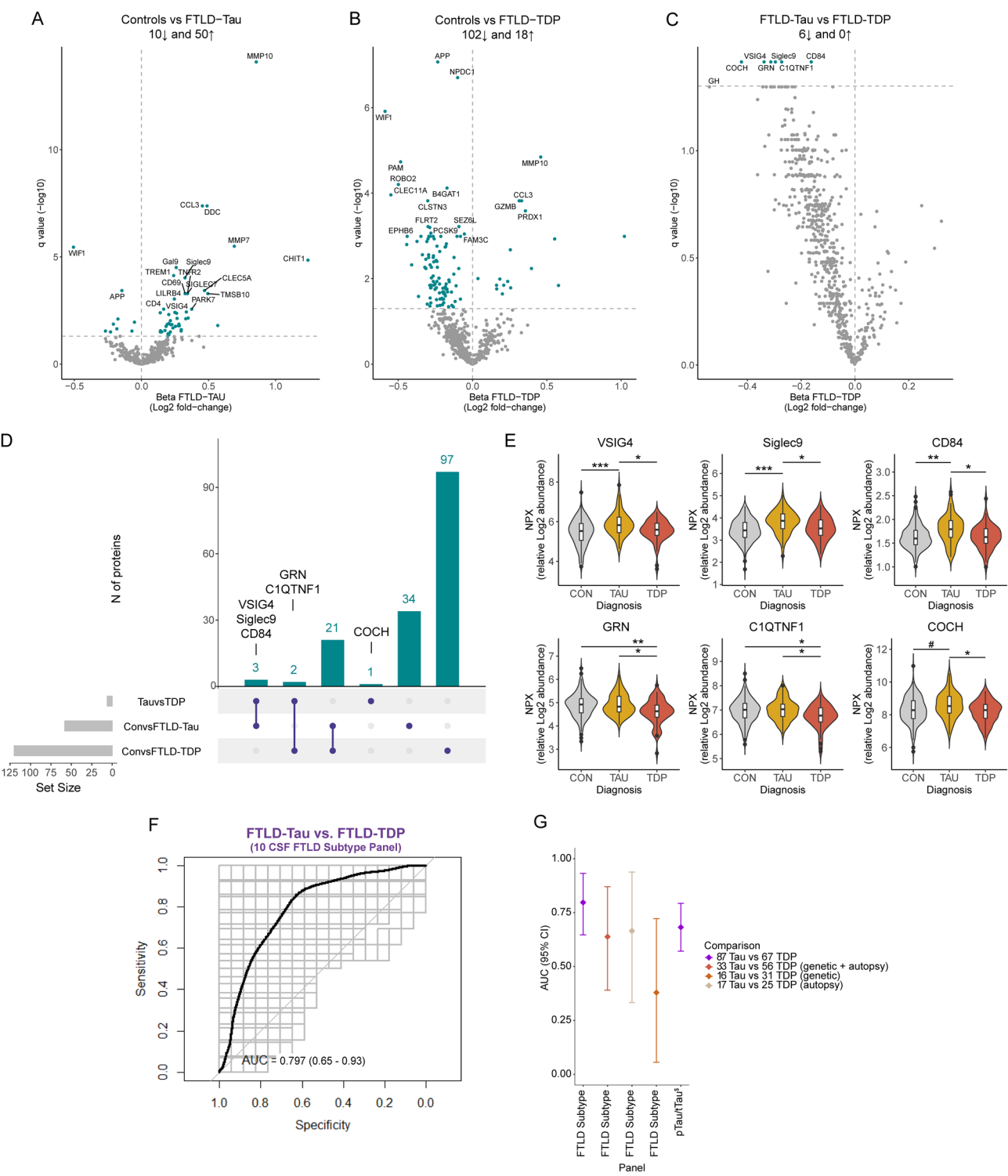


Fig. 3 (See legend on next page.)

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Fig. 3 Differential abundance of CSF proteins in FTLD subtypes and FTLD biomarker panel. Volcano plots show CSF proteins differentially regulated between patients with FTLD-Tau ($n=87$; **A**) or FTLD-TDP ($n=67$; **B**) and controls ($n=196$) and between these neuropathological subtypes (**C**). Each dot represents a protein. The beta coefficients ($\log_2$ fold-change) are plotted versus q values ($-\log_{10}$ -transformed). Proteins significantly dysregulated after adjusting for false discovery rate (FDR, $q < 0.05$) are depicted in blue. The total number of proteins that are down-regulated (left) or up-regulated (right) is indicated. **D**) UpSet plot depicts proteins dysregulated between the FTLD-Tau, FTLD-TDP, and control groups. **E**) Violins represent the abundance ($\log_2$ NPX) of the CSF proteins that were uniquely dysregulated in FTLD-Tau and FTLD-TDP or between the subtypes. Boxplot within the violin indicates the median and interquartile range of the protein abundance. **F**) ROC curves depict the performance of 10-CSF biomarker panel discriminating FTLD-Tau ($n=87$) from FTLD-TDP ($n=67$). **G**) Forest plot shows the different AUC and 95% confidence interval for the 10 CSF biomarker panels, and pTau/tTau ratio to discriminate between neuropathological subtypes (purple; 87 FTLD-Tau and 67 FTLD-TDP), subtypes that are genetic and/or autopsy confirmed (brown; 33 FTLD-Tau and 56 FTLD-TDP), subtypes which are genetically confirmed (coral; 16 FTLD-Tau and 31 FTLD-TDP), and subtypes which are autopsy confirmed (beige; 17 FTLD-Tau and 25 FTLD-TDP). \$ shows the comparison in a subset of cases (NfL: 76 FTD, 47 controls, and 54 AD; pTau/tTau ratio: 74 FTLD-Tau and 37 FTLD-TDP. # $p < 0.05$, * $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$. Abbreviations: CON, cognitively unimpaired controls; TDP, transactive response DNA binding protein 43

are specifically associated with one of the FTLD subtypes, we visualized the outcomes of three comparisons in an upset plot (Fig. 3D). Here, we identified three proteins uniquely related to FTLD-Tau (i.e., all increased in FTLD-Tau compared to controls and FTLD-TDP; Fig. 3E). These proteins play a role in various processes, including phagocytosis and the NLRP3 inflammasome (VSIG4), cell adhesion processes (Siglec9), and involvement in both the innate and adaptive immune responses (CD84) [58–60]. Two proteins were uniquely related to FTLD-TDP (i.e., both decreased in FTLD-TDP compared to controls and FTLD-Tau; Fig. 3E), which are related to lysosomal functioning (GRN) and the immune response (C1QTNF1) [61, 62]. Interestingly, one protein was uniquely dysregulated between FTLD-Tau and FTLD-TDP (COCH; Fig. 3E), with higher levels in FTLD-Tau. This protein plays a role in the modulation of cell shapes [63].

CSF protein panel to discriminate the FTLD subtypes

We next sought to identify CSF biomarkers specific to the FTLD subtypes. A panel of 10 CSF proteins that discriminated between FTLD-Tau and FTLD-TDP with moderate performance was identified (AUC: 0.797, 95% CI: 0.65–0.93; Fig. 3F). Despite the performance being superior to the pTau/tTau ratio (AUC: 0.68, 95% CI: 0.57–0.79; Fig. 3G), the large CI reflects high variability in the cross-validation procedures and thus poor robustness. This model included proteins that were specifically dysregulated in FTLD-Tau (i.e., VSIG4, CD84, and Siglec9) or FTLD-TDP (i.e., GRN) and those that were nominally or significantly changed between subtypes (i.e., COCH, GH, CST5, and PRSS8; Supplementary Fig. 3), as well as proteins that were similar across groups (i.e., IFNLR1 and ANGPT2; Supplementary Fig. 4). The performance of this model did not improve after stratifying the FTLD groups for genetic or sporadic cases (AUCs: 0.37–0.64; Fig. 3G).

Validation of FTD panels using custom PEA assays in independent cohorts

We next aimed to translate our proteomic findings into clinically useful tools. Thus, we developed custom multiplex-PEA assays containing 18 of the 24 proteins from the FTD panels described above (i.e., FTD diagnostic panel: 9 proteins; FTD differential diagnostic panel: 12 proteins; 3 proteins present in both panels). Six proteins could not be included due to technical limitations. The custom PEA assays (custom panels) showed over 90% detectability of the proteins in patient samples, with low coefficients of variation (CVs; <5% intra-CV and <3% inter-CV; Supplementary Table 6). No custom multiplex-PEA assay was developed for the FTLD subtype panel.

We compared the protein's fold changes obtained in the custom panels for three validation cohorts (including one with autopsy confirmation) to those in the discovery cohort. The protein's fold changes obtained in the comparisons between FTD and controls significantly correlated between those detected in the discovery and two validation cohorts (clinical cohort 1: $Rho=0.45$, $p > 0.05$, clinical cohort 2: $Rho=0.65$, $p < 0.05$ and FTLD/AD autopsy cohort: $Rho=0.93$, $p < 0.001$; Supplementary Fig. 5A). In addition, the fold changes of WIF1 and MMP10 were strong and replicated across all cohorts (Fig. 4A). Similarly, strong correlations of the protein fold changes for the difference between FTD and AD patients were observed between the discovery cohort and the three validation cohorts (clinical cohort 1: $Rho=0.69$, $p < 0.05$; clinical cohort 2: $Rho=0.78$, $p < 0.01$; FTLD/AD autopsy cohort: $Rho=0.87$, $p < 0.001$, Supplementary Fig. 5B). Noteworthy, we observed that the fold changes of some proteins lost significance after correction for multiple testing in clinical cohort 1 and clinical cohort 2, whereas several proteins showed their strongest effects in the FTLD/AD autopsy cohort (e.g., THOP1, MMP7, ENO2, CCL11, and KAZALD1; Figs. 4A and 5A).

The FTD diagnostic panel (containing 9 out of the 14 proteins) confirmed discrimination of FTD from controls with high accuracy in all validation cohorts (clinical cohort 1: AUC=0.96, 95% CI: 0.92-1; clinical cohort 2: AUC=0.96, 95% CI: 0.92-1; FTLD/AD autopsy

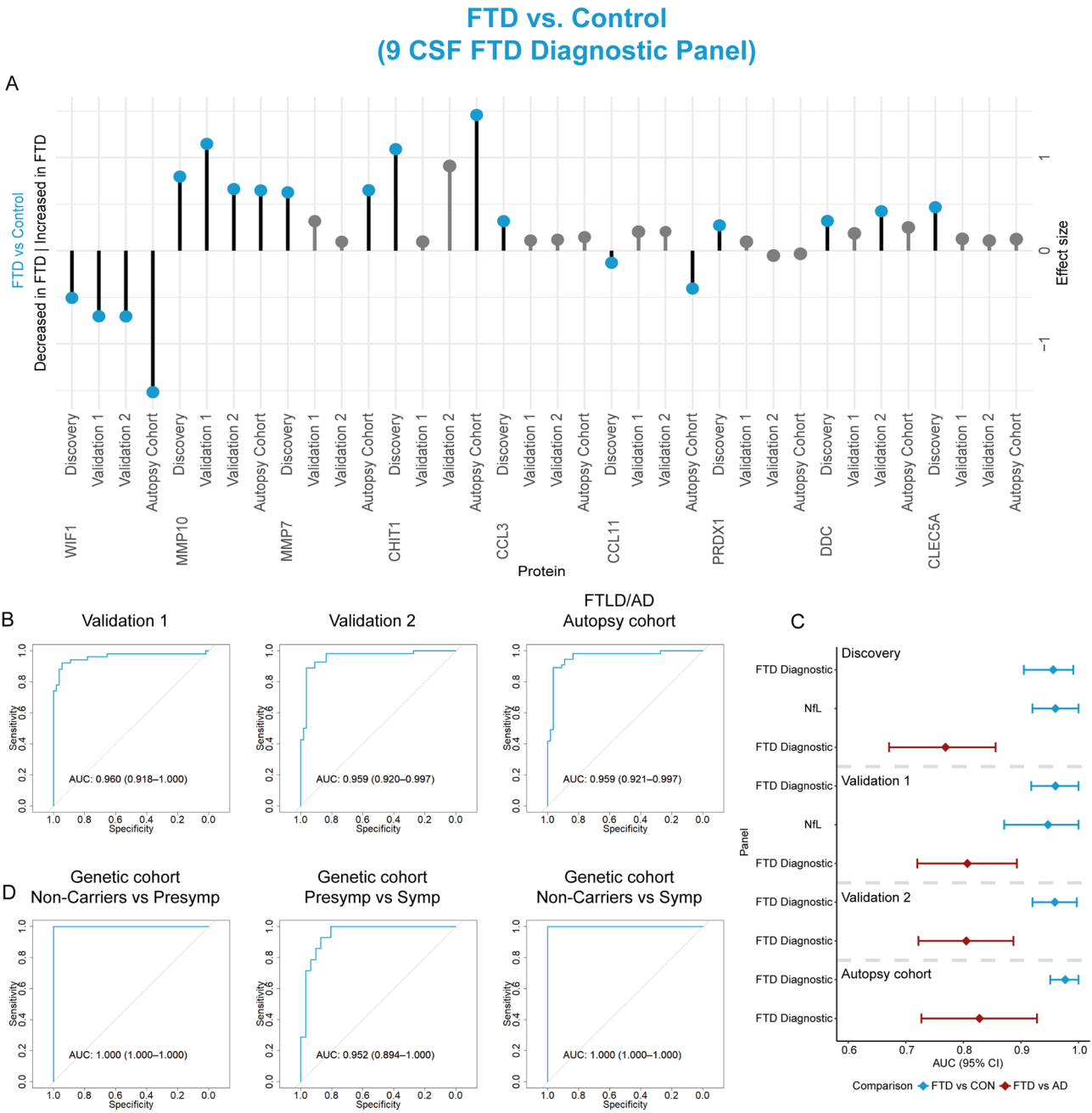


Fig. 4 Development and validation of custom CSF biomarker panels for FTD diagnosis in independent cohorts. **(A)** Lollipop plots depict the beta-coefficients obtained in the discovery phase in parallel to the beta-coefficients of the custom assays in clinical validation cohorts 1 and 2 and the FTLD/AD autopsy cohort. Grey dots show proteins that did not remain significant after correction for multiple testing. **(B)** Receiver operating characteristic (ROC) curves showing the performance of the CSF biomarker panel discriminating FTD from controls using the custom assays across the two clinical and one autopsy validation cohort. Inserts outline corresponding AUC and 95% CI. **(C)** Forest plots depict the different AUC and 95% CI obtained with the CSF FTD biomarker panels or CSF NfL in the comparison between FTD and controls (blue) or AD (red). **(D)** ROC curves showing the performance of the FTD diagnostic panel in the FTD genetic cohort to discriminate non-carriers from presymptomatic (PreSymp) and symptomatic (Symp) mutation carriers, and presymptomatic from symptomatic

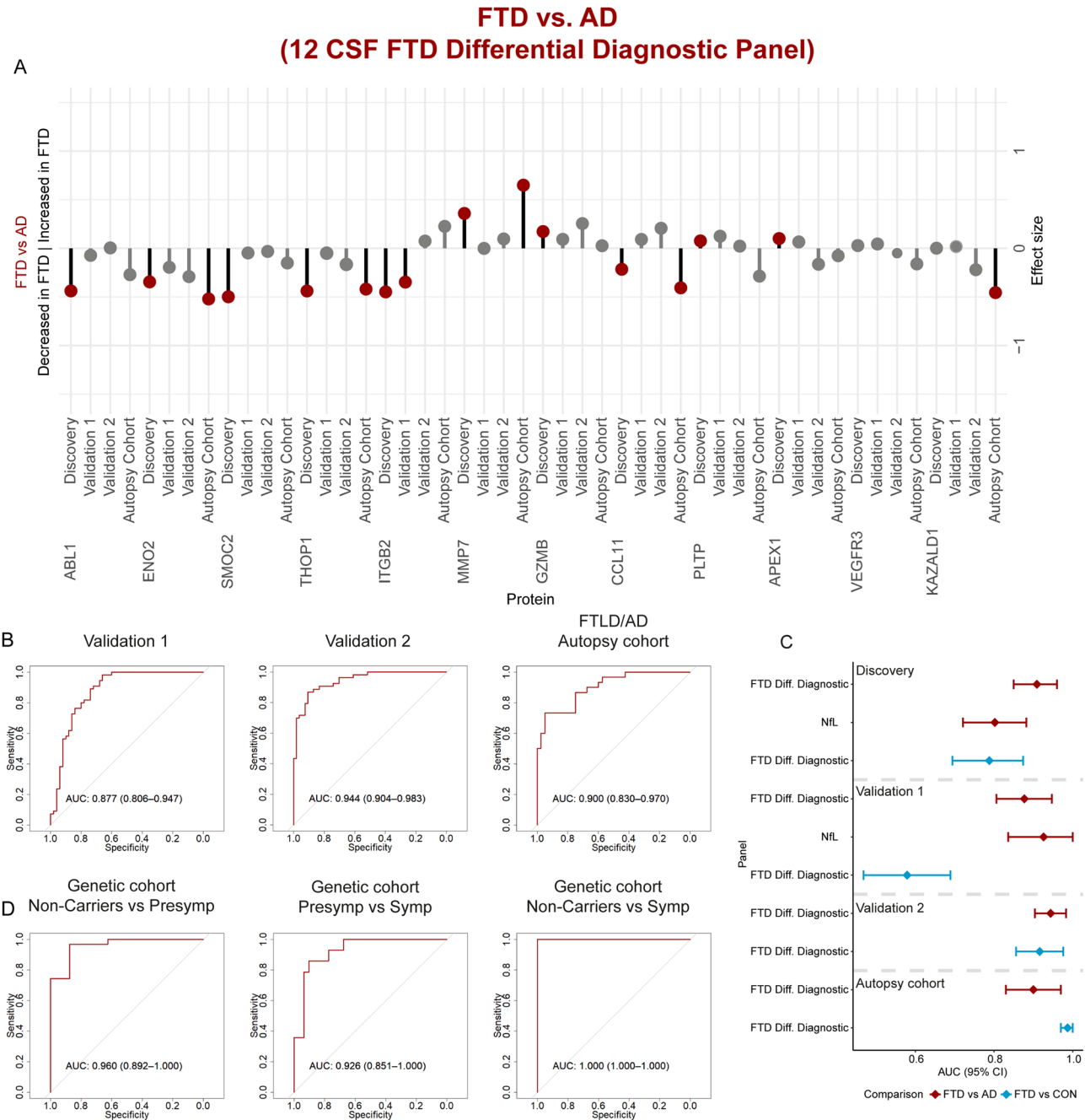


Fig. 5 Development and validation of custom CSF biomarker panels for FTD differential diagnosis in independent cohorts. **(A)** Lollipop plots depict the beta-coefficients obtained in the discovery phase in parallel to the beta-coefficients of the custom assays in clinical validation cohorts 1 and 2 and the FTLD/AD autopsy cohort. Grey dots show proteins that did not remain significant after correction for multiple testing. **(B)** Receiver operating characteristic (ROC) curves showing the performance of the CSF biomarker panel discriminating FTD from AD using the custom assays across the two clinical and one autopsy validation cohort. Inserts outline corresponding AUC and 95% CI. **(C)** Forest plots depict the different AUC and 95% CI obtained with the CSF FTD biomarker panels or CSF NfL in the comparison between FTD and AD (red) or controls (blue). **(D)** ROC curves showing the performance of the FTD differential diagnostic panel in the FTD genetic cohort to discriminate non-carriers from presymptomatic (PreSymp) and symptomatic (Symp) mutation carriers, and presymptomatic from symptomatic

cohort: AUC=0.98, 95% CI: 0.95–1; Fig. 4B) and was similar to CSF NfL (clinical cohort 1: AUC=0.95, 95% CI: 0.87–1; Fig. 4C). Exclusion of the FTD patients with AD co-pathology ($n=8$) in clinical cohort 1 had minimal impact on the performance of the FTD diagnostic panel (AUC=0.960, 95% CI: 0.91–1; Supplementary Fig. 6A). In addition, this FTD diagnostic panel showed strong performance for discriminating symptomatic mutation carriers from non-carriers (AUC=1, 95% CI: 1–1) and presymptomatic carriers (AUC=0.95, 95% CI: 0.90–1) as well as in differentiating presymptomatic carriers from non-carriers (AUC=1, 95% CI: 1–1; Fig. 4D; Supplementary Fig. 7). The FTD differential diagnostic panel (containing 12 out of the 13 proteins) could again discriminate FTD from AD with high accuracy in the three validation cohorts (clinical cohort 1: AUC=0.88, 95% CI: 0.81–0.95; clinical cohort 2: AUC=0.94, 95% CI: 0.90–0.98; FTLD/AD autopsy cohort: AUC=0.90, 95% CI: 0.83–0.97; Fig. 5B), which was similar to NfL (clinical cohort 1, AUC=0.93, 95% CI: 0.84–1; Fig. 5C). Exclusion of the FTD patients with AD co-pathology ($n=8$) in clinical cohort 1 increased the diagnostic performance of the FTD differential diagnostic panel (AUC=0.939, 95% CI: 0.87–0.99; Supplementary Fig. 6B). In addition, the FTD differential diagnostic panel showed good performance for discriminating symptomatic mutation carriers from non-carriers (AUC=1, 95% CI: 1–1) and presymptomatic carriers (AUC=0.93, 95% CI: 0.86–1) as well as in differentiating presymptomatic carriers from non-carriers (AUC=0.96, 95% CI: 0.90–1; Fig. 5D; Supplementary Fig. 8). To summarize, we replicated the discriminative performance as we observed in the discovery cohort, both for the FTD diagnostic and differential diagnostic panel.

Discussion

This study identified novel and specific protein changes for FTD and its main pathological subtypes. These findings were translated into two CSF biomarker panels that discriminated FTD from controls and AD patients with high accuracy (AUCs > 0.9). These results were validated by clinically feasible panels that measured the selected proteins in several independent cohorts, including an autopsy-confirmed and a genetic FTD cohort. We identified proteins specifically associated with FTLD-Tau or FTLD-TDP groups, but we could not identify a marker or combination of markers to robustly discriminate between these FTLD subgroups. Moreover, the panels identified the presymptomatic stage of FTD mutation carriers with high accuracy, while the further increase in the symptomatic stage suggests their potential as an objective tool for monitoring disease progression. The FTD proteins and biomarker panels identified in this study reflect a broad range of different biological processes associated with

FTD and its biological subtypes including inflammatory processes, regulation of synapses, lysosomal functioning, tissue remodeling, and oxidative stress.

Fluid biomarkers specifically associated with FT(L)D pathophysiology are needed to improve diagnostic accuracy, for clinical trial inclusion, and to monitor treatment effects [14]. We have performed a large multicenter FTD proteomics study, including patients with different FTLD neuropathologies as well as a group with AD dementia. Up to 92 proteins were dysregulated between FTD and controls. By comparing the protein profiles to AD patients, we observed that 40% of these proteins were uniquely dysregulated in FTD (e.g., WIF1, ROBO2, and SLITRK2). Some proteins have been associated with FTD in previous CSF antibody-based proteomics (VEGFA [64]), or FTLD brain proteomics studies (ADAM23 [65–67] and WIF1 [68]). We also identified a subset of CSF proteins (30%) that were differentially regulated in both patients with FTD and AD compared to controls but with a different protein abundance between FTD and AD. Despite most of these being more prominently dysregulated in AD (e.g., SDC4, ITGB2, MIF, and sTREM1) [29, 69, 70], a small subset of proteins showed an opposite effect between these dementias (e.g., CHL1, GPC1, and CNTN5). Previous studies additionally detected decreased levels of CNTN5 and CHL1 in FTD and PSP using alternative platforms, which further supports the validity of our findings [71–73]. In addition, we detected a subset of proteins (20%) likely reflecting an overall neurodegenerative process, as they were similarly dysregulated in AD and FTD compared with controls. For instance, CHIT1 and MMP10 protein levels were increased in FTD and AD, as reported previously [29, 64, 74–79]. This study identifies numerous CSF proteins dysregulated in FTD. However, due to the inclusion of an AD group, we uncovered that half (43 out of 92) of the proteins are specifically associated with FTD, highlighting the importance of comparing to both controls and similar neurological diseases in biomarker studies. Among the proteins specifically associated with FTD, the ones showing the strongest effects were: WIF1, MMP7, APP, NPDC1, and IL1RL2. WIF1 is an inhibitory protein involved in the Wnt signaling cascade, a dysregulation of this pathway has been implicated in tau phosphorylation and other neuronal processes (e.g., neurogenesis, synaptic health and plasticity) [80, 81]. In addition, WIF1 was dysregulated in FTLD frontal cortex tissue [68], supporting its role within FTD pathophysiology, however, its specific function in relation to FTD remains to be elucidated. Enrichment analysis showed that proteins specifically dysregulated in FTD were enriched in biological pathways associated with axon development, regulation of synapse assembly, and cell-cell adhesion mediator activity. This is in line with previous unbiased brain and CSF

proteomics studies, and multiplex CSF analysis showing a dysregulation of similar pathways in FTLT [31, 65, 66, 82, 83].

Considering that biomarkers to discriminate the FTLT-subtypes are currently lacking. We additionally investigated the CSF proteome in the main FTLT pathological subtypes. Despite many CSF proteins that were dysregulated in FTLT-Tau (60 proteins) or FTLT-TDP (120 proteins) compared to controls, overlapping analysis across all comparisons revealed three proteins related to FTLT-Tau (VSIG4, Siglec9, and CD84), two proteins related to FTLT-TDP (GRN and C1QTNF1), and one protein uniquely changed between the neuropathological subtypes (COCH). Given the established association between *GRN* mutations and TDP pathology, the decrease of GRN (progranulin) protein levels observed in FTLT-TDP aligns with previous GRN reductions in CSF and plasma of FTD-*GRN* mutation carriers [84, 85]. It should be noted that the FTLT-TDP group included a low number of *GRN* mutation carriers (11%). Considering that we observed decreased GRN levels in the total FTLT-TDP group, these results highlight that lysosomal dysfunction is an important feature in TDP pathology, which is additionally supported by post-mortem analysis [86, 87]. Furthermore, we also identified novel markers such as VSIG4. This protein was also found to be elevated in *MAPT* mutation carriers compared to non-carriers in a recent MS-proteomics study [88]. As a phagocytic receptor, it plays a key role in inflammatory responses and can mediate activation of the NLRP3 inflammasome [58]. Since the NLRP3 inflammasome plays a role within tau pathophysiology, this protein seems promising to reflect ongoing activation of the NLRP3 inflammasome in FTD patients [89]. To the best of our knowledge, many of the subtype-specific proteins identified in this study are novel and have not yet been linked to the main FTD pathologies or genetic subtypes in previous proteomics studies [30, 31, 65, 66]. Thus, these proteins require further investigation in CSF and brain tissue, to examine their relevance to ongoing tau or TDP43 pathophysiology.

Next, we identified a classification signature of 10 CSF proteins to discriminate the two main FTLT subtypes (AUC: 0.8), with higher performance than the pTau/tTau ratio (AUC: 0.68). However, the moderate performance together with the large confidence intervals suggests that the model is rather unstable and may vary depending on the samples included. This variation could be explained by the inclusion of different FTD biochemical profiles, which likely differ between sporadic and familial cases with the same proteopathy, or even between genetic backgrounds of the same pathological subtypes (e.g., *GRN* and *C9orf72*) [14, 71, 88, 90]. Despite the unprecedented number of FTLT samples and proteins

analyzed in this study, we identified relatively few robust markers associated with specific pathological subtypes. This limited discriminative power underscores both the biological complexity of FTLT and the inherent difficulty of distinguishing subtypes in vivo, a challenge that remains a critical hurdle for the field. While previous studies have reported promising results in differentiating broader diagnostic categories such as AD and FTD, the finer stratification of the FTLT subtypes requires deeper molecular insights. Notably, we observed a number of proteins with nominally significant differences across the FTLT subtypes, suggesting that larger, more powered studies may reveal additional subtype-specific biomarkers. These findings also reflect the heterogeneity within each FTLT subtype, emphasizing the importance of analyzing well-characterized cohorts, ideally with pathological and/or genetic confirmed individuals [14].

To translate the CSF proteome findings into practical biomarker tools for routine diagnostics or clinical trials, we applied classification analyses and identified two panels of 14 and 13 CSF proteins that can discriminate FTD from controls and AD dementia with high accuracies (AUCs of 0.96 and 0.91, respectively). Thereafter, custom multiplex assays were developed and validated in several independent cohorts. The fold changes for each protein correlated well between discovery and validation cohorts (*Rho*'s between 0.45 and 0.93). The high discriminative values for FTD were confirmed in three cohorts for both panels (AUCs > 0.877), supporting the robustness of our findings. In addition, these panels show added value by identifying FTD mutation carriers approaching symptom onset, and could thus be relevant to detect early pathological changes. Noteworthy, our novel FTD panels have similar discriminative performance compared to NfL. However, while NfL reflects neuroaxonal damage, our CSF panels offer a more comprehensive depiction of the underlying biological processes in FTD. The relevance of these FTD panels is additionally supported by the association of some markers (e.g., MMP10, PLTP, PRDX1, and NPDC1) with clinical parameters such as cognitive functioning or disease severity. For example, we detected positive correlations of APEX, PRDX1, and CHIT1 with FTLT-CDR scores. These proteins are related to oxidative stress and reactive microglial activity, processes that underlie FTD pathogenesis, and may therefore be relevant for disease staging [91, 92]. In addition, in agreement with previous studies, we observed that increased MMP10 levels were associated with worse cognitive performance [29, 74]. This could potentially be explained by the expression of MMP10 by microglial cells in the brain, as elevated MMP10 levels contribute to ongoing inflammatory responses leading to axonal damage, potentially leading to functional deficits such as cognitive impairment [93]. Overall, these associations provide further

support for the potential value of these panels in clinical settings and trial contexts.

This study is not without limitations. The targeted proteomic approach employed in this study misses relevant proteins that might have been measured using unbiased MS methods (e.g., GFAP and neuropentraxins). However, our study still covered >600 proteins covering a wide range of biological mechanisms and the workflow employed allowed us to swiftly translate our proteomic discovery findings into custom immunoassays for subsequent validation. Noteworthy, potential FTD misdiagnosis may influence biomarker results. However, diagnoses were made in specialized memory clinics and 47% of FTD individuals were either pathologically or genetically confirmed, and results were further validated in multiple independent cohorts. Furthermore, we observed that a subset of confirmed FTD patients in the discovery cohort exhibited a positive AD biomarker profile indicative of AD-co-pathology. Exclusion of these cases minimally affected the differential protein expression analyses, but increased the discriminative potential in the validation cohorts, which can probably be attributed to the association of several proteins in the panel with AD pathology (e.g., ABL1, THOP1, SMOC2, and ITGB2). Nevertheless, the potential influence of AD co-pathology on the FTD proteome should be explored in future studies involving larger sample sizes. Noteworthy, validation in the FTD genetic cohort demonstrated a high discriminative performance of our CSF biomarker panels in both presymptomatic and symptomatic FTD carriers compared to non-carriers. Although the smaller sample size in this cohort, relative to our other cohorts, may impact the reliability of the analysis, the strong performance of the panel remains promising for detecting FTD across different disease stages. However, these biomarker panels are not yet ready for clinical implementation. Additional validation in independent FTD cohorts is required to confirm their robustness and generalizability. Furthermore, it would be valuable to assess the performance of these novel panels in other disorders that share clinical and/or neuropathological features with FTD, such as primary psychiatric disorders or amyotrophic lateral sclerosis.

Conclusions

This study identified CSF proteome changes specifically associated to FTD. We have translated these findings into CSF biomarker panels that can accurately discriminate FTD from controls and AD in multiple cohorts, and in the presymptomatic stage of the disease. Furthermore, we detect protein changes specifically associated to FTLT-Tau or FTD-TDP, although larger and more homogeneous cohorts will have more power to discriminate between these main pathological subtypes. The panels developed within this study could prove valuable

for diagnosis or to monitor treatment effects in clinical trials. The antibody-based technology employed in this study allowed us to efficiently translate our discovery findings into custom multiplex panels for further clinical validation showing reproducible findings. This workflow applied here could also be helpful for the development of fluid biomarkers in other human matrices (e.g., blood) and other biomedical fields beyond neurodegenerative dementias.

Abbreviations

AD	Alzheimer's dementia
ADC	Amsterdam dementia cohort
AUC	Area under the curve
CSF	Cerebrospinal fluid
CON	Cognitively normal individuals
CBS	Corticobasal syndrome
FDR	False discovery rate
FTD	Frontotemporal dementia
FTLD	Frontal temporal lobar degeneration
LOD	Lower limit of detection
MMSE	Mini-mental state examination
MS	Mass spectrometry
NPX	Normalized protein expression
PEA	Proximity extension assay
PPA	Primary progressive aphasia
PSP	Progressive supranuclear
QC	Quality controls

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13024-025-00882-5>.

Supplementary Material 1
Supplementary Material 2
Supplementary Material 3
Supplementary Material 4
Supplementary Material 5
Supplementary Material 6
Supplementary Material 7

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Not applicable.

Author contributions

Y.H., M.C., and C.T. conceived and designed the study. Y.H., L.V., and C.P. performed the statistical analysis. Y.H., L.V., E.E., S.B., L.M., J.S., J.H., H.R., A.L., D.A., S.E., A.S., A.C.-P., D.I., W.F., Y.P., M.C., and C.T. recruited participants and collected clinical data and samples. Y.H. and M.C. arranged and prepared samples for proteomics analysis. Y.H., M.C., and C.T. drafted the manuscript. All authors contributed to the revision and editing of the manuscript.

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Data availability

The source data generated in this study are available within this study (Supplementary Table 2, Supplementary Table 3, Supplementary Table 4, Supplementary Table 5). All models were built using publicly available packages and functions in R.

Declarations

Ethics approval and consent to participate

Approval was given by the institutional ethical review boards of each center. Written informed consent was obtained from all participants or their authorized representatives.

Consent for publication

Not applicable.

Competing interests

Y.H., C.P., E.E., S.B., L.M., J.S., J.H., H.R., W.H., A.S., A.C.-P. and D.I. declare no competing interest. M.C. has been an invited speaker at Eisai, is an associate editor at Alzheimer's Research & Therapy and has been an invited writer for Springer Healthcare. L.V. received a grant for CORAL consortium by Olink proteomics. D.I. is a Scientific Advisory Board Member for Denali Therapeutics. D.A. participated in advisory boards from Fujirebio-Europe and Roche Diagnostics and received speaker honoraria from Fujirebio-Europe, Roche Diagnostics, Nutricia, Krka Farmacéutica S.L., Zambon S.A.U. and Esteve Pharmaceuticals. S.A. D.A., and A.L. declare a filed patent application (Title: Markers of synaptopathy in neurodegenerative disease; Applicant: Fundació Institut de Recerca de l'Hospital de la Santa Creu i Sant Pau, Inventors: Olívia BELBIN; Alberto LLEÓ; Alejandro BAYÉS; Juan FORTEA; Daniel ALCOLEA; Application number: PCT/EP2019/056535; International Publication Number: WO 2019/175379 A1; Current status: Active. Licensed to ADx Neurociences NV (Ghent, Belgium); this patent is not related to any specific aspect of the current manuscript). S.E. received personal fees from Eisai (paid to institution), personal fees from icometrix (paid to institution), personal fees from Novartis (paid to institution), personal fees from Roche (paid to institution) and personal fees from Roche and personal fees from Biogen, all outside the submitted work. W.F. has performed contract research for Biogen MA Inc, and Boehringer Ingelheim. W.F. has been an invited speaker at Boehringer Ingelheim, Biogen MA Inc, Danone, Eisai, WebMD Neurology (Medscape), Springer Healthcare. W.F. is consultant to Oxford Health Policy Forum CIC, Roche, and Biogen MA Inc. W.F. participated in advisory boards of Biogen MA Inc and Roche. All funding is paid to her institution. W.F. is a member of the steering committee of PAVE, and Think Brain Health. W.F. was associate editor of Alzheimer's Research & Therapy in 2020/2021 and is currently an associate editor at Brain. C.E.T. has a collaboration contract with ADx Neurociences, Quanterix, and Eli Lilly, performed contract research or received grants from AC-Immune, Axon Neurosciences, Bioconnect, Bioorchestra, Brainstorm Therapeutics, Celgene, EIP Pharma, Eisai, Grifols, Novo Nordisk, PeopleBio, Roche, Toyama, and Vivoryon. She serves on editorial boards of *Medicines*, *Neurology*/Springer, *Alzheimer's Research & Therapy*, *Neurology*, *Neuroimmunology & Neuroinflammation*, and is editor of a *Neuromethods* book Springer. She had speaker contracts for Roche, Grifols, and Novo Nordisk.

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References

1. Gorno-Tempini ML, Hillis AE, Weintraub S, Kertesz A, Mendez M, Cappa SF et al. Classification of primary progressive aphasia and its variants. *Am Acad Neurol*. 2011;76.
2. Rascovsky K, Hodges JR, Knopman D, Mendez MF, Kramer JH, Neuhaus J, et al. Sensitivity of revised diagnostic criteria for the behavioural variant of frontotemporal dementia. *Brain*. 2011;134(Pt 9):2456–77.
3. Armstrong MJ, Litvan I, Lang AE, Bak TH, Bhatia KP, Borroni B et al. Criteria for the diagnosis of corticobasal degeneration. *Am Acad Neurol*. 2013;80.
4. Hoglinger GU, Respondek G, Stamelou M, Kurz C, Josephs KA, Lang AE, et al. Clinical diagnosis of progressive supranuclear palsy: the movement disorder society criteria. *Mov Disord*. 2017;32(6):853–64.
5. Irwin DJ, Cairns NJ, Grossman M, McMillan CT, Lee EB, Van Deerlin VM, et al. Frontotemporal lobar degeneration: defining phenotypic diversity through personalized medicine. *Acta Neuropathol*. 2015;129(4):469–91.
6. Mann DMA, Snowden JS. Frontotemporal lobar degeneration: pathogenesis, pathology and pathways to phenotype. *Brain Pathol*. 2017;27(6):723–36.
7. Neumann M, Rademakers R, Roeber S, Baker M, Kretschmar HA, Mackenzie IR. A new subtype of frontotemporal lobar degeneration with FUS pathology. *Brain*. 2009;132(Pt 11):2922–31.
8. Lashley T, Rohrer JD, Mead S, Revesz T. Review: an update on clinical, genetic and pathological aspects of frontotemporal lobar degenerations. *Neuropathol Appl Neurobiol*. 2015;41(7):858–81.
9. Rohrer JD, Guerreiro R, Vandrovicova J, Uphill J, Reiman D, Beck J et al. The heritability and genetics of frontotemporal lobar degeneration. *Neurology*. 2009;73.
10. Greaves CV, Rohrer JD. An update on genetic frontotemporal dementia. *J Neurol*. 2019;266(8):2075–86.
11. Pozzi FE, Aprea V, Giovannelli G, Lattuada F, Crivellaro C, Bertola F, et al. Clinical and neuroimaging characterization of the first frontotemporal dementia family carrying the MAPT p.K298E mutation. *Neurogenetics*. 2024;25(3):215–23.
12. Cheng HR, Lin RR, Li HL, Xue YY, Gao PR, Chen DF, et al. Identification and functional characterization of novel variants of MAPT and GRN in Chinese patients with frontotemporal dementia. *Neurobiol Aging*. 2023;123:233–43.
13. Neumann M, Lee EB, Mackenzie IR. Frontotemporal lobar degeneration TDP-43-immunoreactive pathological subtypes: clinical and mechanistic significance. *Adv Exp Med Biol*. 2021;1281:201–17.
14. Del Campo M, Zetterberg H, Gandy S, Onyike CU, Oliveira F, Udeh-Momoh C, et al. New developments of biofluid-based biomarkers for routine diagnosis and disease trajectories in frontotemporal dementia. *Alzheimers Dement*. 2022;18(11):2292–307.
15. Hutchinson AD, Mathias JL. Neuropsychological deficits in frontotemporal dementia and alzheimer's disease: a meta-analytic review. *J Neurol Neurosurg Psychiatry*. 2007;78(9):917–28.

16. Naasan G, Rabinovici GD, Ghosh P, Eloffson JD, Miller BL, Coppola G, et al. Amyloid in dementia associated with Familial FTL: not an innocent bystander. *Neurocase*. 2016;22(1):76–83.
17. Irwin DJ, Trojanowski JQ, Grossman M. Cerebrospinal fluid biomarkers for differentiation of frontotemporal lobar degeneration from alzheimer's disease. *Front Aging Neurosci*. 2013;5:6.
18. Ashton NJ, Janelidze S, Al Khleifat A, Leuzy A, van der Ende EL, Karikari TK, et al. A multicentre validation study of the diagnostic value of plasma neurofilament light. *Nat Commun*. 2021;12(1):3400.
19. Bridel C, van Wieringen WN, Zetterberg H, Tijms BM, Teunissen CE, and the, NFLG et al. Diagnostic value of cerebrospinal fluid neurofilament light protein in neurology: a systematic review and meta-analysis. *JAMA Neurol*. 2019;76(9):10354–8.
20. Ducharme S, Dols A, Laforce R, Devenney E, Kumfor F, van den Stock J, et al. Recommendations to distinguish behavioural variant frontotemporal dementia from psychiatric disorders. *Brain*. 2020;143(6):1632–50.
21. Gendron TF, Heckman MG, White LJ, Veire AM, Pedraza O, Burch AR, et al. Comprehensive cross-sectional and longitudinal analyses of plasma neurofilament light across FTD spectrum disorders. *Cell Rep Med*. 2022;3(4):100607.
22. Meeter LHH, Vijverberg EG, Del Campo M, Rozemuller AJM, Donker Kaat L, de Jong FJ, et al. Clinical value of neurofilament and phospho-tau/tau ratio in the frontotemporal dementia spectrum. *Neurology*. 2018;90(14):e1231–9.
23. Pijnenburg YA, Verwey NA, van der Flier WM, Scheltens P, Teunissen CE. Discriminative and prognostic potential of cerebrospinal fluid phosphotau/tau ratio and neurofilaments for frontotemporal dementia subtypes. *Alzheimers Dement (Amst)*. 2015;1(4):505–12.
24. Cousins KAQ, Shaw LM, Chen-Plotkin A, Wolk DA, Van Deerlin VM, Lee EB, et al. Distinguishing frontotemporal lobar degeneration Tau from TDP-43 using plasma biomarkers. *JAMA Neurol*. 2022;79(11):1155–64.
25. Desmarais P, Rohrer JD, Nguyen QD, Herrmann N, Stuss DT, Lang AE, et al. Therapeutic trial design for frontotemporal dementia and related disorders. *J Neurol Neurosurg Psychiatry*. 2019;90(4):412–23.
26. Higginbotham L, Ping L, Dammer EB, Duong DM, Zhou M, Gearing M, et al. Integrated proteomics reveals brain-based cerebrospinal fluid biomarkers in asymptomatic and symptomatic alzheimer's disease. *Sci Adv*. 2020;6:43.
27. Bader JM, Geyer PE, Muller JB, Strauss MT, Koch M, Leyboldt F, et al. Proteome profiling in cerebrospinal fluid reveals novel biomarkers of alzheimer's disease. *Mol Syst Biol*. 2020;16(6):e9356.
28. Dammer EB, Ping L, Duong DM, Modeste ES, Seyfried NT, Lah JJ, et al. Multi-platform proteomic analysis of alzheimer's disease cerebrospinal fluid and plasma reveals network biomarkers associated with proteostasis and the matrisome. *Alzheimers Res Ther*. 2022;14(1):174.
29. Del Campo M, Peeters CFW, Johnson ECB, Vermunt L, Hok-A-Hin YS, van Nee M, et al. CSF proteome profiling across the alzheimer's disease spectrum reflects the multifactorial nature of the disease and identifies specific biomarker panels. *Nat Aging*. 2022;2(11):1040–53.
30. Teunissen CE, Elias N, Koel-Simmelink MJ, Durieux-Lu S, Malekzadeh A, Pham TV, et al. Novel diagnostic cerebrospinal fluid biomarkers for pathologic subtypes of frontotemporal dementia identified by proteomics. *Alzheimers Dement (Amst)*. 2016;2:86–94.
31. van der Ende EL, Meeter LH, Stingl C, van Rooij JGJ, Stoop MP, Nijholt DAT, et al. Novel CSF biomarkers in genetic frontotemporal dementia identified by proteomics. *Ann Clin Transl Neurol*. 2019;6(4):698–707.
32. Del Campo M, Galimberti D, Elias N, Boonkamp L, Pijnenburg YA, van Swieten JC, et al. Novel CSF biomarkers to discriminate FTL and its pathological subtypes. *Ann Clin Transl Neurol*. 2018;5(10):1163–75.
33. van der Ende EL, Xiao M, Xu D, Poos JM, Panman JL, Jiskoot LC, et al. Neuronal pentraxin 2: a synapse-derived CSF biomarker in genetic frontotemporal dementia. *J Neurol Neurosurg Psychiatry*. 2020;91(6):612–21.
34. Bolsewig K, Hok AHYS, Sepe FN, Boonkamp L, Jacobs D, Bellomo G, et al. A combination of neurofilament light, glial fibrillary acidic protein, and neuronal Pentraxin-2 discriminates between frontotemporal dementia and other dementias. *J Alzheimers Dis*. 2022;90(1):363–80.
35. Llorens F, Thune K, Tahir W, Kanata E, Diaz-Lucena D, Xanthopoulos K, et al. YKL-40 in the brain and cerebrospinal fluid of neurodegenerative dementias. *Mol Neurodegener*. 2017;12(1):83.
36. Del Campo M, Vermunt L, Peeters CFW, Sieben A, Hok AHYS, Lleo A, et al. CSF proteome profiling reveals biomarkers to discriminate dementia with lewy bodies from alzheimer's disease. *Nat Commun*. 2023;14(1):5635.
37. van der Flier WM, Scheltens P. Amsterdam dementia cohort: performing research to optimize care. *J Alzheimers Dis*. 2018;62(3):1091–111.
38. Ingelsson M, Ramasamy K, Russ C, Freeman SH, Orne J, Raju S, et al. Increase in the relative expression of Tau with four microtubule binding repeat regions in frontotemporal lobar degeneration and progressive supranuclear palsy brains. *Acta Neuropathol*. 2007;114(5):471–9.
39. Alcolea D, Clarimon J, Carmona-Iragui M, Illan-Gala I, Morenas-Rodriguez E, Barroeta I, et al. The Sant Pau initiative on neurodegeneration (SPIN) cohort: A data set for biomarker discovery and validation in neurodegenerative disorders. *Alzheimers Dement (N Y)*. 2019;5:597–609.
40. Dubois B, Feldman HH, Jacova C, Dekosky ST, Barberger-Gateau P, Cummings J, et al. Research criteria for the diagnosis of alzheimer's disease: revising the NINCDS-ADRDA criteria. *Lancet Neurol*. 2007;6(8):734–46.
41. Cairns NJ, Bigio EH, Mackenzie IR, Neumann M, Lee VM, Hatanpaa KJ, et al. Neuropathologic diagnostic and nosologic criteria for frontotemporal lobar degeneration: consensus of the consortium for frontotemporal lobar degeneration. *Acta Neuropathol*. 2007;114(1):5–22.
42. Montine TJ, Phelps CH, Beach TG, Bigio EH, Cairns NJ, Dickson DW, et al. National institute on aging-alzheimer's association guidelines for the neuropathologic assessment of alzheimer's disease: a practical approach. *Acta Neuropathol*. 2012;123(1):1–11.
43. Miyagawa T, Brushaber D, Syrjanen J, Kremers W, Fields J, Forsberg LK, et al. Utility of the global CDR plus NACC FTL rating and development of scoring rules: data from the ARTFL/LEFFTDS consortium. *Alzheimers Dement*. 2020;16(1):106–17.
44. Hok-A-Hin YS, Willemse EAJ, Teunissen CE, Del Campo M. Guidelines for CSF processing and biobanking: impact on the identification and development of optimal CSF protein biomarkers. *Methods Mol Biol*. 2019;2044:27–50.
45. Tijms BM, Willemse EAJ, Zwan MD, Mulder SD, Visser PJ, van Berckel BNM, et al. Unbiased approach to counteract upward drift in cerebrospinal fluid Amyloid-beta 1–42 analysis results. *Clin Chem*. 2018;64(3):576–85.
46. Vermunt L, Otte M, Verberk IMW, Killestein J, Lemstra AW, van der Flier WM, et al. Age- and disease-specific reference values for neurofilament light presented in an online interactive support interface. *Ann Clin Transl Neurol*. 2022;9(11):1832–7.
47. Proteomics O. Development and validation of customized PEA biomarker panels with clinical utility.
48. Benjamini Y, Hochberg Y. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *J Royal Stat Soc Ser B (Methodological)*. 1995;57.
49. de Leeuw FA, Peeters CFW, Kester MI, Harms AC, Struys EA, Hankemeier T, et al. Blood-based metabolic signatures in alzheimer's disease. *Alzheimers Dement (Amst)*. 2017;8:196–207.
50. Friedman J, Hastie T, Tibshirani R. Regularization paths for generalized linear models via coordinate descent. *J Stat Softw*. 2010;33(1):22.
51. Zhou Y, Zhou B, Pache L, Chang M, Khodabakhshi AH, Tanaseichuk O, et al. Metascape provides a biologist-oriented resource for the analysis of systems-level datasets. *Nat Commun*. 2019;10(1):1523.
52. Karlsson L, Vogel J, Arvidsson I, Åström K, Janelidze S, Blennow K et al. Cerebrospinal fluid reference proteins increase accuracy and interpretability of biomarkers for brain diseases. *BioRxiv*. 2023.
53. Poe M, Blake JT, Boulton DA, Gammon M, Sigal NH, Wu JK, et al. Human cytotoxic lymphocyte granzyme B. Its purification from granules and the characterization of substrate and inhibitor specificity. *J Biol Chem*. 1991;266(1):98–103.
54. Mroczko B, Groblewska M, Barcikowska M. The role of matrix metalloproteinases and tissue inhibitors of metalloproteinases in the pathophysiology of neurodegeneration: a literature study. *J Alzheimers Dis*. 2013;37(2):273–83.
55. Spencer ML, Theodosiou M, Noonan DJ. NPDC-1, a novel regulator of neuronal proliferation, is degraded by the ubiquitin/proteasome system through a PEST degradation motif. *J Biol Chem*. 2004;279(35):37069–78.
56. Pei DS, Jia PP, Luo JJ, Liu W, Strauss PR. AP endonuclease 1 (Apex1) influences brain development linking oxidative stress and DNA repair. *Cell Death Dis*. 2019;10(5):348.
57. Olsson AK, Dimberg A, Kreuger J, Claesson-Welsh L. VEGF receptor signalling - in control of vascular function. *Nat Rev Mol Cell Biol*. 2006;7(5):359–71.
58. Huang X, Feng Z, Jiang Y, Li J, Xiang Q, Guo S et al. VSIG4 mediates transcriptional inhibition of Nlrp3 and Il-1b in macrophages. *Sci Adv*. 2019;5.
59. Zhang JQ, Nicoll G, Jones C, Crocker PR. Siglec-9, a novel Sialic acid binding member of the Immunoglobulin superfamily expressed broadly on human blood leukocytes. *J Biol Chem*. 2000;275(29):22121–6.
60. Yan Q, Malashkevich VN, Fedorov A, Fedorov E, Cao E, Lary JW, et al. Structure of CD84 provides insight into SLAM family function. *PNAS*. 2007;104:25.

61. Tanaka Y, Suzuki G, Matsuaki T, Hosokawa M, Serrano G, Beach TG, et al. Progranulin regulates lysosomal function and biogenesis through acidification of lysosomes. *Hum Mol Genet*. 2017;26(5):969–88.
62. Kishore U, Gaboriaud C, Waters P, Shrive AK, Greenhough TJ, Reid KB, et al. C1q and tumor necrosis factor superfamily: modularity and versatility. *Trends Immunol*. 2004;25(10):551–61.
63. Goel M, Sienkiewicz AE, Picciani R, Lee RK, Bhattacharya SK. Cochlin induced TREK-1 co-expression and Annexin A2 secretion: role in trabecular meshwork cell elongation and motility. *PLoS ONE*. 2011;6(8):e23070.
64. Bostrom G, Freyhult E, Virhammar J, Alcolea D, Tumani H, Otto M, et al. Diferent inflammatory signatures in alzheimer's disease and frontotemporal dementia cerebrospinal fluid. *J Alzheimers Dis*. 2021;81(2):629–40.
65. Mol MO, Miedema SSM, Melhem S, Li KW, Koopmans F, Seelaar H, et al. Proteomics of the dentate gyrus reveals semantic dementia specific molecular pathology. *Acta Neuropathol Commun*. 2022;10(1):190.
66. Miedema SSM, Mol MO, Koopmans FTW, Hondius DC, van Nierop P, Menden K, et al. Distinct cell type-specific protein signatures in GRN and MAPT genetic subtypes of frontotemporal dementia. *Acta Neuropathol Commun*. 2022;10(1):100.
67. Umoh ME, Dammer EB, Dai J, Duong DM, Lah JJ, Levey AI, et al. A proteomic network approach across the ALS-FTD disease spectrum resolves clinical phenotypes and genetic vulnerability in human brain. *EMBO Mol Med*. 2018;10(1):48–62.
68. Mofrad RB, Del Campo M, Peeters CFW, Meeter LHH, Seelaar H, Koel-Simmlink M, et al. Plasma proteome profiling identifies changes associated to AD but not to FTD. *Acta Neuropathol Commun*. 2022;10(1):148.
69. Hok-A-Hin YS, Del Campo M, Boiten WA, Stoops E, Vanhooren M, Lemstra AW, et al. Neuroinflammatory CSF biomarkers MIF, sTREM1, and sTREM2 show dynamic expression profiles in alzheimer's disease. *J Neuroinflammation*. 2023;20(1):107.
70. Zhang S, Zhao J, Zhang Y, Zhang Y, Cai F, Wang L, et al. Upregulation of MIF as a defense mechanism and a biomarker of alzheimer's disease. *Alzheimers Res Ther*. 2019;11(1):54.
71. Remnestal J, Oijerstedt L, Ullgren A, Olofsson J, Bergstrom S, Kultima K, et al. Altered levels of CSF proteins in patients with FTD, presymptomatic mutation carriers and non-carriers. *Transl Neurodegener*. 2020;9(1):27.
72. Wise A, Li J, Yamakawa M, Loureiro J, Peterson B, Worringer K, et al. CSF proteomics in patients with progressive supranuclear palsy. *Neurology*. 2024;103(3):e209585.
73. Muqaku B, Anderl-Straub S, Werner L, Nagl M, Otto M, Teunissen CE, et al. Contactin proteins in cerebrospinal fluid show different alterations in dementias. *J Neurol*. 2024;271(12):7516–24.
74. Martino Adami PV, Orellana A, Garcia P, Kleineidam L, Alarcon-Martin E, Montreuil L, et al. Matrix metalloproteinase 10 is linked to the risk of progression to dementia of the alzheimer's type. *Brain*. 2022;145(7):2507–17.
75. Abu-Rumeileh S, Steinacker P, Polisch B, Mammana A, Bartoletti-Stella A, Oeckl P, et al. CSF biomarkers of neuroinflammation in distinct forms and subtypes of neurodegenerative dementia. *Alzheimers Res Ther*. 2019;12(1):2.
76. Steinacker P, Verde F, Fang L, Feneberg E, Oeckl P, Roeber S, et al. Chitotriosidase (CHIT1) is increased in microglia and macrophages in spinal cord of amyotrophic lateral sclerosis and cerebrospinal fluid levels correlate with disease severity and progression. *J Neurol Neurosurg Psychiatry*. 2018;89(3):239–47.
77. van der Ende EL, Veld S, Hanskamp I, van der Lee S, Dijkstra JIR, Hok AHYS, CSF proteomics in autosomal dominant Alzheimer's disease highlights parallels with sporadic disease. *Brain*. 2023.
78. Whelan CD, Mattsson N, Nagle MW, Vijayaraghavan S, Hyde C, Janelidze S, et al. Multiplex proteomics identifies novel CSF and plasma biomarkers of early alzheimer's disease. *Acta Neuropathol Commun*. 2019;7(1):169.
79. Guo Y, Chen SD, You J, Huang SY, Chen YL, Zhang Y et al. Multiplex cerebrospinal fluid proteomics identifies biomarkers for diagnosis and prediction of Alzheimer's disease. *Nat Hum Behav*. 2024.
80. Bovolenta P, Esteve P, Ruiz JM, Cisneros E, Lopez-Rios J. Beyond Wnt inhibition: new functions of secreted Frizzled-related proteins in development and disease. *J Cell Sci*. 2008;121(Pt 6):737–46.
81. Hadi F, Akrami H, Shahpasand K, Fattahi MR. Wnt signalling pathway and Tau phosphorylation: A comprehensive study on known connections. *Cell Biochem Funct*. 2020;38(6):686–94.
82. Sogorb-Esteve A, Nilsson J, Swift IJ, Heller C, Bocchetta M, Russell LL, et al. Differential impairment of cerebrospinal fluid synaptic biomarkers in the genetic forms of frontotemporal dementia. *Alzheimers Res Ther*. 2022;14(1):118.
83. Saloner R. Large-scale network analysis of the cerebrospinal fluid proteome identifies molecular signatures of frontotemporal Lobar degeneration. Preprint *Biorxiv*. 2024.
84. Goossens J, Bjerke M, Van Mossevelde S, Van den Bossche T, Goeman J, De Vil B, et al. Diagnostic value of cerebrospinal fluid tau, neurofilament, and progranulin in definite frontotemporal Lobar degeneration. *Alzheimers Res Ther*. 2018;10(1):31.
85. Meda F, Simren J, Borroni B, Cantoni V, Archetti S, Biasiotto G et al. Analytical and clinical validation of a blood progranulin ELISA in frontotemporal dementias. *Clin Chem Lab Med*. 2023.
86. Davis SE, Cook AK, Hall JA, Voskobiynik Y, Carullo NV, Boyle NR, et al. Patients with sporadic FTL exhibit similar increases in lysosomal proteins and storage material as patients with FTD due to GRN mutations. *Acta Neuropathol Commun*. 2023;11(1):70.
87. Mackenzie IR, Baker M, Pickering-Brown S, Hsiung GY, Lindholm C, Dwosh E, et al. The neuropathology of frontotemporal Lobar degeneration caused by mutations in the progranulin gene. *Brain*. 2006;129(Pt 11):3081–90.
88. Sogorb-Esteve A, Weiner S, Simren J, Swift IJ, Bocchetta M, Todd EG, et al. Proteomic analysis reveals distinct cerebrospinal fluid signatures across genetic frontotemporal dementia subtypes. *Sci Transl Med*. 2025;17(784):eadm9654.
89. Ising C, Venegas C, Zhang S, Scheiblich H, Schmidt SV, Vieira-Saecker A, et al. NLRP3 inflammasome activation drives Tau pathology. *Nature*. 2019;575(7784):669–73.
90. Lleo A, Irwin DJ, Illan-Gala I, McMillan CT, Wolk DA, Lee EB, et al. A 2-Step cerebrospinal algorithm for the selection of frontotemporal Lobar degeneration subtypes. *JAMA Neurol*. 2018;75(6):738–45.
91. Palluzzi F, Ferrari R, Graziano F, Novelli V, Rossi G, Galimberti D, et al. A novel network analysis approach reveals DNA damage, oxidative stress and calcium/camp homeostasis-associated biomarkers in frontotemporal dementia. *PLoS ONE*. 2017;12(10):e0185797.
92. Malpetti M, Cope TE, Street D, Jones PS, Hezemans FH, Mak E, et al. Microglial activation in the frontal cortex predicts cognitive decline in frontotemporal dementia. *Brain*. 2023;146(8):3221–31.
93. Nuttall RK, Silva C, Hader W, Bar-Or A, Patel KD, Edwards DR, et al. Metalloproteinases are enriched in microglia compared with leukocytes and they regulate cytokine levels in activated microglia. *Glia*. 2007;55(5):516–26.

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